

Alex Nikolaenko

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PROFILE

Dual BSc in Mathematics and Economics (HSE Moscow · University of London International Programme), heading to London for an MSc in Financial Mathematics in 2026. Comfortable across stochastic calculus, time-series econometrics, and the engineering side of quantitative work — from short-rate calibration to building agentic LLM workflows.

EDUCATION

BSc Mathematics & Economics (*Dual Degree*)

2022 — 2026

HSE Moscow · University of London International Programme

GPA 9.24/10 · top 0.6% of cohort

Secondary School

2015 — 2022

Moscow School No. 57 — Mathematics and Economics class

EXPERIENCE

Junior Business Analyst

Feb — Apr 2026

T-Bank

- Translated business requirements into engineering specifications; coordinated between product stakeholders and development teams.
- Built practical foundation in system architecture and project management within a large-bank tech environment.

Student Mentor & Tutor

2025 — 2026

HSE Moscow

- Onboarding and year-round support for incoming first-year students.
- Voluntary Macroeconomics II tutoring sessions for peers.

SELECTED PROJECTS

CS:GO / CS2 Skin Price Analysis

Python · ARMA–GARCH · Cointegration

Modelled log-price dynamics of in-game cosmetics; identified volatility clustering, fat tails, and seasonality. Applied Zivot–Andrews structural-break detection and Engle–Granger cointegration to find co-moving skin pairs — demonstrating statistical parallels with traditional financial assets.

MOEX G-Curve Modelling

Python · Short-rate Models · EVT/GPD · MLE

Calibrated Vasicek, Cox–Ingersoll–Ross, and Hull–White short-rate models to Russian government bond data via maximum likelihood. Fitted return-distribution tails with the Generalised Pareto Distribution; assessed goodness-of-fit through empirical-vs-theoretical quantile coverage.

SKILLS

Programming Python · R · SQL · $\text{\LaTeX}$

Tools Git · Pandas · NumPy · SciPy · PyTorch · Docker

Quant & Finance Stochastic calculus · Time series · Linear algebra · Risk modelling

AI & Agents Claude Code · Anthropic SDK · MCP server integration · Multi-agent orchestration

LANGUAGES

Russian — native | English — fluent